PROJECT PROPOSAL

Road Accident Severity Prediction Using Machine Learning

Virtual Data Science Apprentice Program

Python Specialist Intern — Week 1 Task: Project Planning and Dataset Scoping

Program Duration: 10-Aug-2026 to 07-Sep-2026

Submission Date: 12-Aug-2026

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1. Executive Summary

Road traffic accidents remain a leading cause of injury and death worldwide, and understanding the factors that drive accident severity is essential for prevention, emergency-response planning, and infrastructure investment. This proposal outlines a data science project that applies machine learning in Python to predict the severity of road accidents based on environmental, temporal, and road-condition factors. The project uses a large, publicly available accident dataset and follows a structured workflow spanning data acquisition, cleaning, exploratory analysis, feature engineering, model building, and evaluation. The goal is to produce a reliable classification model together with clear, actionable insights that could support road-safety decision-making.

2. Project Objectives and Goals

The project is guided by the following objectives:

- **Primary objective:** Build a machine learning model in Python that predicts the severity of a road accident (e.g., Slight, Serious, Fatal, or an equivalent numeric severity scale) using features available at or near the time of the incident.
- **Analytical objective:** Identify which environmental, temporal, and road-related factors (weather, visibility, time of day, junction type, speed limit, etc.) are most strongly associated with higher-severity outcomes.
- **Technical objective:** Demonstrate a complete, reproducible Python-based data science pipeline — from raw data to a validated, evaluated model — using industry-standard libraries.
- **Communication objective:** Translate model outputs and exploratory findings into visualizations and a narrative that a non-technical stakeholder (e.g., a transport authority) could act on.

Success will be measured by: a cleaned and well-documented dataset; a trained classification model evaluated with accuracy, precision, recall, and F1-score (with particular attention to recall on the most severe class); and a set of clearly communicated, data-backed insights.

3. Project Scope

To keep the project realistic within the internship timeline, the scope is deliberately bounded:

- **In scope:** Structured, tabular accident-record data; supervised classification of severity; standard tabular ML models (e.g., logistic regression, decision trees, random forest, gradient boosting); exploratory data analysis and visualization; model evaluation using held-out test data.
- **Out of scope:** Real-time prediction or deployment as a live service; use of unstructured data such as accident photos or police narrative text; causal inference (the project identifies association, not proven causation); collection of new primary data.
- **Geographic / temporal scope:** Analysis will be based on the selected public dataset's coverage area and time period; findings will be explicitly framed as applying to that scope rather than generalized globally.

4. Anticipated Challenges

Several challenges are anticipated based on the nature of accident-severity data, and are addressed in more detail in the Risk Register (Section 9):

- Class imbalance, since fatal and serious accidents are much rarer than minor ones.
- Missing or inconsistent values in weather, lighting, and road-surface fields.
- Mixed data types (categorical, numerical, datetime, geospatial) requiring careful preprocessing.

- Dataset size, which may require sampling or efficient data-loading strategies.
- Risk of misleading conclusions if correlation is mistaken for causation.

5. Dataset Selection

Three candidate public datasets were reviewed for suitability. The comparison below summarizes the rationale for the final selection.

Dataset	Source	Size	Suitability
US Accidents (2016–2023)	Kaggle (Sobhan Moosavi et al.)	~7.7M records, 46 columns	Selected — rich features, severity label included
UK Road Safety Data	UK Dept. for Transport (data.gov.uk)	~1.5M records/year	Alternative — strong quality, narrower geography
RTA Dataset (Addis Ababa)	Kaggle / UCI mirror	~12,000 records	Backup — small, useful if compute is limited

5.1 Selected Dataset: US Accidents (2016–2023)

The US Accidents dataset (compiled by Sobhan Moosavi and collaborators, hosted on Kaggle) was selected as the primary data source. It aggregates traffic accident records across the United States using multiple data providers, including traffic APIs that stream incident data from state and local transportation departments, law-enforcement agencies, and traffic cameras.

Key reasons for selection:

- Scale: several million records provide enough data for robust model training and for the minority (severe) classes to still contain a meaningful number of examples.
- Rich feature set: includes weather conditions, visibility, temperature, precipitation, time of day, road features (junction, crossing, traffic signal), and a severity label — directly matching the project's predictive target.
- Public availability and license: freely available for research and educational use, with clear documentation of fields and collection methodology.
- Realistic messiness: the dataset contains missing values and inconsistencies typical of real-world data, making it a valuable exercise in practical data cleaning rather than a pre-cleaned toy dataset.

The UK Road Safety dataset is noted as a strong alternative if a narrower, government-curated dataset is preferred, and the smaller RTA (Addis Ababa) dataset is kept as a lightweight backup in case computational resources become a constraint.

6. Analytical Workflow

The project follows a linear, checkpoint-driven workflow from raw data to final reporting:

- **Step 1 — Data Acquisition:** Download the dataset, verify file integrity, and load it into a pandas DataFrame; confirm row counts, column names, and data types.
- **Step 2 — Data Cleaning:** Identify and handle missing values, duplicate records, and outliers; standardize categorical labels and datetime formats (detailed further in the Week 2 task).
- **Step 3 — Exploratory Data Analysis:** Examine distributions of severity, time, weather, and location features; visualize relationships and correlations to form hypotheses (detailed further in the Week 3 task).

- **Step 4 — Feature Engineering:** Encode categorical variables, extract time-based features (hour, day of week, season), and scale numerical features.
- **Step 5 — Model Selection and Training:** Train and compare candidate classification models, using cross-validation to tune hyperparameters (detailed further in the Week 4 task).
- **Step 6 — Evaluation:** Assess models using accuracy, precision, recall, F1-score, and a confusion matrix, with emphasis on correctly identifying severe cases.
- **Step 7 — Reporting:** Summarize findings, visualizations, and model performance into a clear narrative for stakeholders.

6.1 Workflow Diagram

Data Acquisition → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Model Selection & Training → Evaluation → Reporting

(Each arrow represents a checkpoint where outputs are validated before moving to the next stage, as outlined in Section 8.)

7. Methodology and Tools

The project will be implemented entirely in Python, using the following tools at each stage of the workflow:

Stage	Python Library / Tool	Purpose
Data Acquisition	pandas, kagglehub / requests	Download and load raw CSV data into DataFrames
Data Cleaning	pandas, numpy	Handle missing values, duplicates, type conversion
Exploratory Analysis	matplotlib, seaborn, plotly	Visualize distributions, correlations, geographic patterns
Feature Engineering	scikit-learn, pandas	Encode categoricals, scale features, derive time-based features
Modeling	scikit-learn, XGBoost	Train classification models to predict severity class
Evaluation	scikit-learn.metrics	Compute accuracy, precision, recall, F1, confusion matrix
Reporting	Jupyter Notebook, Matplotlib	Document findings with narrative and visuals

Candidate models for the classification task include logistic regression (as an interpretable baseline), decision trees and random forests (for handling non-linear relationships and mixed feature types), and gradient-boosted trees such as XGBoost (for higher predictive performance). Model choice will be finalized in Week 4 based on a systematic comparison against the evaluation metrics defined above.

8. Project Timeline and Checkpoints

The Week 1 task is planned across seven days, with a validation checkpoint at the end of each day to keep the proposal on track:

Day(s)	Milestone / Checkpoint	Validation Criteria
Day 1	Problem statement finalized; candidate datasets shortlisted	Problem statement is specific and measurable
Day 2	Dataset selected and downloaded; initial structure reviewed	Dataset loads correctly; column list and row count confirmed
Day 3	Project objectives, scope, and success metrics documented	Objectives are SMART and tied to evaluation metrics
Day 4	High-level workflow and tool stack drafted	Every workflow stage maps to a specific Python library
Day 5	Risk register and mitigation plan written	Each identified risk has a paired mitigation step
Day 6	Full proposal drafted end-to-end	Document contains all required sections
Day 7	Review, proofreading, and formatting; final submission	Document passes self-review checklist and is exported to DOCX

9. Risk Register and Mitigation Plan

The table below documents the main risks anticipated for this project and the corresponding mitigation strategy for each.

Challenge	Potential Impact	Mitigation Strategy
Severe class imbalance (fatal accidents are rare)	Model biased toward majority class, poor recall on severe cases	Use stratified sampling, SMOTE, class-weighted loss functions
Large dataset size	Slow processing on local hardware	Work with a stratified subsample; use chunked loading in pandas
Missing / inconsistent weather and road-condition fields	Reduced feature quality and reliability	Apply domain-informed imputation; drop fields below a completeness threshold
Geographic and temporal bias in source data	Findings may not generalize beyond the source region/period	State scope limitations explicitly; avoid overgeneralized claims

10. Conclusion

This proposal establishes a clear, achievable plan for building a road-accident severity prediction project in Python. By selecting a large, feature-rich public dataset, defining measurable objectives, and laying out a structured workflow with explicit checkpoints, the project is positioned to move smoothly into the Week 2 data-cleaning phase. The plan balances technical ambition — using real-world, imperfect data and a genuine classification problem — with realism about scope, timeline, and known data-quality risks, providing a solid foundation for the remaining weeks of the internship.